

TRAMASURE-100

(Tramadol Hydrochloride Injection 50mg/ml)

PRODUCT NAME:

TRAMASURE-100 (Tramadol Hydrochloride Injection 50mg/ml)

NAME AND STRENGTH OF ACTIVE INGREDIENTS:

TRAMASURE-100: Each 2 mL contains Tramadol Hydrochloride Ph. Eur. 100 mg and Water for injection USP qs.

PRODUCT DESCRIPTION:

Before reconstitution: Clear and colorless solution.

After reconstitution with infusion solutions:

Clear and colorless solution when mixed with below diluents.

- 0.9% Sodium Chloride Intravenous Infusion
- 5% Glucose Intravenous infusion
- Sodium Lactate Solution

Light brownish color solution when mixed with below diluent

- Haemaccel

PHARMACOLOGICAL PROPERTIES

Pharmacotherapeutic group: Other opioids; ATC-code N02AX02.

Pharmacodynamics Properties

Tramadol is a centrally acting opioid analgesic. It is a nonselective pure agonist at micro, δ - and κ -opioid receptors with a higher affinity to the micro receptor. Other mechanisms, which may contribute to its analgesic effect, are inhibition of neuronal re-uptake of noradrenaline and enhancement of serotonin release.

Tramadol possesses antitussive properties. At variance to morphine, tramadol administered at therapeutic doses has a negligible depressive effect on respiration. Equally, gastrointestinal motility is affected to a lesser extent. Effects on the cardiovascular system are rather small.

The analgesic potency of tramadol is stated to be 1/10 to 1/6 of that of morphine. The onset of the analgesic effect of tramadol is rapid and lasts for several hours (4-6 hrs).

Pharmacokinetics Properties

More than 90% of tramadol is absorbed after oral administration. The half-life of the distribution phase $t_{1/2a}$ is about 0.8 h. Tramadol and its metabolites are eliminated via the kidneys. The half-life of the terminal elimination phase $t_{1/2b}$ in human is in favourable range for analgesics (about 6 h). The biological half-lives of the metabolites are the same range as the non-metabolized substance. Consequently, on repeated administration, therapeutic complications due to slowly eliminated metabolites are unlikely.

The absolute bioavailability of tramadol is about 68% on oral administration and about 79% on rectal administration. The difference between absorbed and available non-metabolized.

Tramadol after oral administration is probably due to the low first-pass effect.

INDICATIONS: Moderate to severe pain.

RECOMMENDED DOSAGE:

The dose should be adjusted to the intensity of the pain and the sensitiveness of each patient.

Adults and adolescents (12 years and older):

IV: 100mg or up to 400mg daily, to be administered by slow injection or diluted in solution for administration by infusion. Infusion solutions that can be prepared with **TRAMASURE-100** are mentioned as below:-

- 0.9% Sodium Chloride Intravenous Infusion
- 5% Glucose Intravenous infusion
- Sodium Lactate Solution
- Haemaccel

The mixture should be prepared the same day the solution is to be used

IM/SC: 100mg

Total daily dose: up to 400mg/day

If the pain relief is still insufficient 30-60 min after administration of 50-100 mg, a 2nd single dose of 50 mg may be used. If the required amount for analgesic is expected to be higher, in cases of severe pain conditions, the initial dose can be increased based on clinical experience. The lowest effective analgesic dose should generally be chosen. The total daily dose of 400 mg should not be exceeded, except in special circumstances.

Tramadol is not approved for use in patients below 12 years old.

Paediatric population:

The safety and efficacy of tramadol has not been studied in the paediatric population. Therefore, use of tramadol is not recommended in patients under 12 years of age.

Special dosage:

In general, the daily dose should not exceed 400mg of tramadol. In case of impaired renal or hepatic function it may be necessary to adjust the dose. For geriatric patients above 75 years of age, it is recommended the dosage should not exceed 300mg daily. The recommended daily dose in renal failure is 200mg and cirrhosis is 50mg every 12 hours.

Duration of the treatment: In case of long term treatment with tramadol the possibility of dependence cannot be excluded. Treatment breaks or alternative therapy should be considered. This product should not be given for longer time than therapeutically necessary.

Route of Administration: For IV, IM & SC Use Only

For single use only.

Do not use if solution is not clear or has suspended matter. Discard any unused solution.

CONTRAINDICATIONS:

1. Hypersensitivity to tramadol.
2. Acute intoxication by alcohol, hypnotics, analgesics, opioid or psychotropic drugs.

3. Patients who are receiving MAO inhibitors (including selegiline) or have used them during the last 14 days.
4. Patients suffering from epilepsy, in case it is not sufficiently controlled by treatment.
5. Tramadol must not be used as a substitute in opioid-dependent patients. Although Tramadol is an opioid, it does not suppress morphine withdrawal symptoms.
6. Children younger than 18 years to treat pain after surgery to remove the tonsils and/or adenoids.
7. Adolescents between 12 and 18 years who are obese or have conditions such as obstructive sleep apnea or severe lung disease, which may increase the risk of serious breathing problems.

WARNING AND PRECAUTIONS

1. Tramadol must be used with caution in patients who are addicted to opioids or suffering from head injury, shock, disturbances of consciousness of unclear origin, disturbances of the respiratory center and respiratory function, as well as conditions with increased intracranial pressure.
2. Caution must be exercised when using tramadol in patients with hypersensitivity to opioids.
3. Seizures have been observed in patients taking the recommended dose of tramadol. There may be an increased risk if the recommended daily dose (400 mg) is exceeded. Patients who suffer from epilepsy or are susceptible to seizures should only be treated with tramadol in urgent exceptional cases. Cerebral seizures have been observed in patients using tramadol. These seizures occurred predominantly when other drugs (which reduce the seizure threshold or may have adrenergic effects on the CNS) e.g., tricyclic antidepressants, neuroleptics, MAO inhibitors and serotonin re-uptake inhibitors were taken concomitantly.
4. Tramadol has a slight potential for dependence. On prolonged use, tolerance as well as psychological and physical dependence may occur. In patients who are inclined to drug abuse or dependence on medication, treatment with tramadol is therefore to be limited to short-term use under strict medical supervision. Tramadol is not a suitable substitute drug in the case of opiate dependency. Although tramadol is an opioid agonist, it does not suppress morphine withdrawal symptoms.
5. In patients with a history of opioid addiction, relapses were observed when tramadol was used.
6. In case of IV administration, tramadol must be injected slowly.
7. **Effects on the Ability to Drive or Operate Machinery:** Even when used in accordance with the instructions, tramadol may change reactivity (e.g., causing sleepiness and dizziness) to such an extent that the ability to drive or operate machinery is adversely affected. This is particularly true in combination with alcohol or other psychotropic substances.
8. **Paediatric population:** The safety and efficacy of **TRAMASURE-100** has not been studied in the paediatric population. Therefore, use of **TRAMASURE-100** is not recommended in patients under 12 years of age.
9. **Respiratory depression:** Administer tramadol cautiously in patients for respiratory depression, including patients with substantially decreased respiratory reserve, hypoxia,

hypercapnia, or pre-existing respiratory depression, as in these patients, even therapeutic doses of tramadol may decrease respiratory drive to the point of apnea. In these patients, alternative non-opioid analgesics should be considered. When large doses of tramadol are administered with anaesthetic medications or alcohol, respiratory depression may result. Respiratory depression should be treated as an overdose. If naloxone is to be administered, use cautiously because it may precipitate seizures.

10. **Cytochromes P450 (CYP) 2D6 Ultra-Rapid Metabolism:** Some individuals may be CYP2D6 ultra-rapid metabolisers. These individuals convert tramadol more rapidly than other people into its more potent opioid metabolites O-desmethyltramadol (M1). This rapid conversion could result in higher than expected opioid-like side effects including life-threatening respiratory depression. The prevalence of this CYP2D6 phenotype varies widely and has been estimated at 0.5 to 1% in Chinese, Japanese and Hispanics, 1 to 10% in Caucasians, 3% in African Americans, and 16-28% in North Africans, Ethiopians, and Arabs. Data are not available for other ethnic groups.

11. **Risk from Concomitant Use with Benzodiazepines:** Profound sedation, respiratory depression, coma, and death may result from the concomitant use of tramadol, with benzodiazepines. Observational Studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lower effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid prescribe a lower initial dose of benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when tramadol is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines.

12. **Serotonin Syndrome with Concomitant Use of Serotonergic Drugs:** Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Tramadol with serotonergic drugs. This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal. The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue Tramadol if serotonin syndrome is suspected.

13. *Adrenal Insufficiency:* Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.
14. *Sexual Function / Reproduction:* Long term use of opioids may be associated with decreased sex hormone and levels and symptoms such as low libido, erectile dysfunction or infertility.

INTERACTIONS WITH OTHER MEDICAMENTS:

1. The concomitant administration of tramadol with substances which likewise have a depressant effect on the central nervous system, including alcohol, may potentiate the central effects.
2. If tramadol is eg, combined with barbiturates, prolongation of anaesthesia was observed in animal experiments.
3. The combination of tramadol with a tranquiliser can be expected to result in a favourable effect on the sensation of pain.
4. Clinically relevant interactions due to concomitant or previous administration of cimetidine are unlikely to occur.
5. Concomitant or previous administration of carbamazepine may lead to a reduced analgesic effect and a shorter duration of action.
6. If tramadol and coumarin derivatives (eg, warfarin) are concomitantly administered, the patient must be carefully monitored, as reduced Quick's values with major bleedings and ecchymoses have been observed in some patients.
7. Other CYP3A inhibiting substances eg, ketoconazole and erythromycin can inhibit both the metabolism of tramadol (N-demethylation) and possibly that of the active O-demethylated metabolites. The clinical significance of this interaction is not known.
8. Tramadol can trigger seizures and increase the convulsant potential of SSRIs, tricyclic antidepressants, antipsychotics and other drugs which lower the seizure threshold.
9. Tramadol must not be combined with MAO inhibitors. Life-threatening interactions that affected the CNS as well as respiratory and circulatory functions have been observed when MAO inhibitors were taken within 14 days of administering the opioid pethidine. The same interactions as with MAO inhibitors cannot be ruled out with tramadol.
10. A limited number of studies have shown that the pre- and postoperative administration of the antiemetic 5-HT₃ antagonist, ondansetron, increased the need for tramadol in patients with postoperative pain.

11. Mixed agonist/antagonists (eg, buprenorphine, nalbuphine, pentazocine) in combination with tramadol are not recommended. There is a theoretical possibility that the analgesic effect of a pure agonist may be reduced under such conditions.
12. *Benzodiazepines*: Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death. The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists. Reserve concomitant prescribing of these drugs for use in patients whom alternative treatment options are inadequate. Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.
13. *Serotonergic Drugs*: The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe patient, particularly during treatment initiation and dose adjustment. Discontinue Tramadol if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue).

PREGNANCY AND LACTATION:

Pregnancy

Tramadol has been shown to cross the placenta. There are no adequate and well-controlled studies in pregnant women. Safe use in pregnancy has not been established. **TRAMASURE-100** is not recommended for pregnant women.

Lactation

Approximately 0.1% of the maternal dose of tramadol is excreted in breast milk. In the immediate post-partum period, for maternal oral daily dosage up to 400mg, this corresponds to a mean amount of tramadol ingested by breast-fed infants of 3% of the maternal weight-adjusted dosage. For this reason tramadol should not be used during lactation or alternatively, breast-feeding should be discontinued during treatment with tramadol. Discontinuation of breast-feeding is generally not necessary following a single dose of tramadol.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

Even when used in accordance with the instructions, tramadol may change reactivity (e.g., causing sleepiness and dizziness) to such an extent that the ability to drive or operate machinery is adversely affected. This is particularly true in combination with alcohol or other psychotropic substances.

This medicine can impair cognitive function and can affect a patient's ability to drive safely. When prescribing this medicine, patients should be told:

- The medicine is likely to affect your ability to drive.
- Do not drive until you know how the medicine affects you.
- It is an offence to drive while under the influence of this medicine.

However, you would not be committing an offence if:

- The medicine has been prescribed to treat a medical or dental problem and
- You have taken it according to the instructions given by the prescriber and in the information provided with the medicine and
- It was not affecting your ability to drive safely.

UNDESIRABLE EFFECTS:

The most common side effects are nausea and dizziness, both occurring in >10% of patients.

Nervous System Disorders:

Dizziness, headache, muzziness, changes in appetite, paraesthesia, tremor, respiratory depression, epileptic seizures, involuntary muscle twitching, coordination disorders, syncope.

If the recommended dosage is considerably exceeded and other substances with a centrally depressing action are taken concomitantly, respiratory depression can occur.

Epileptic seizures have occurred predominantly after using high doses of tramadol or after concomitant use of medicines which may have the effect of lowering the seizure threshold.

Psychiatric Disorders:

Hallucinations, confusion, sleep disorders, anxiety and nightmares.

Psychiatric side effects may occur following administration of tramadol, which vary individually in intensity and nature (depending on personality and duration of medication).

These include changes in mood (usually elation, occasionally dysphoria), changes in activity (usually suppression, occasionally increase) and changes in cognitive and sensorial ability (e.g., decision behaviour, perception disorders).

Dependence may occur.

Eye Disorders: Blurred vision.

Cardiac and Vascular Disorders: Effects on cardiovascular regulation (palpitation, tachycardia, orthostatic hypotension or cardiovascular collapse). These undesirable effects may occur especially after IV administration and in patients who are physically stressed. Bradycardia, increased blood pressure.

Respiratory, Thoracic and Mediastinal Disorders: Dyspnoea. Respiratory depression (rare).

Worsening of asthma has been reported. However, no causal relationship could be established.

Gastrointestinal Disorders: Nausea, vomiting, constipation, dry mouth, retching, gastrointestinal irritation (e.g., stomach pressure, bloating), diarrhoea.

Hepatobiliary Disorders: Increases in liver enzyme values have been reported in a temporal connection with the therapeutic use of tramadol.

Skin and Subcutaneous Tissue Disorders: Sweating, dermal reactions (e.g., pruritus, rash, urticaria). There have been serious skin reactions eg, Stevens-Johnson syndrome and toxic epidermal necrolysis in temporal connection with the use of tramadol. In most cases, those affected were patients with multimorbidity undergoing multidrug therapy.

Musculoskeletal and Connective Tissue Disorders: Muscle Weakness.

Renal and Urinary Disorders: Micturition disorders (difficulty or pain in passing urine and urinary retention).

General and Administration Site Conditions: Fatigue, allergic reactions (e.g., dyspnoea, bronchospasm, wheezing, angioneurotic oedema) and anaphylaxis.

Symptoms of withdrawal reactions, similar to those occurring during opiate withdrawal, may occur as follows: Agitation, anxiety, nervousness, insomnia, hyperkinesia, tremor and gastrointestinal symptoms.

Other symptoms which have been observed on very rare occasions when tramadol is discontinued are panic attacks, severe anxiety, hallucinations, paraesthesia, tinnitus and unusual CNS symptoms.

In postmarketing experience, there were reported cases such as follows:

- Serotonin syndrome
- Adrenal insufficiency
- Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility.

The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

- Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

OVERDOSE AND TREATMENT:

Symptoms

In principle, on intoxication with tramadol symptoms similar to those of other centrally acting analgesics (opioids) are to be expected. These include in particular miosis, vomiting, cardiovascular collapse, consciousness disorders up to coma, convulsions and respiratory depression up to respiratory arrest.

Treatment

The general emergency measures apply. Keep open the respiratory tract (aspiration!), maintain respiration and circulation depending on the symptoms. The antidote for respiratory depression is naloxone. In animal experiments naloxone had no effect on convulsions. In such cases diazepam should be given intravenously.

In case of intoxication orally, gastrointestinal decontamination with activated charcoal or by gastric lavage is only recommended within 2 hours after tramadol intake. Gastrointestinal decontamination at a later time point may be useful in case of intoxication with exceptionally large quantities.

Tramadol is minimally eliminated from the serum by haemodialysis or haemo-filtration. Therefore treatment of acute intoxication with Tramadol Hydrochloride Injection with haemodialysis or haemofiltration alone is not suitable for detoxification.

INCOMPATIBILITIES:

Precipitation will occur if Tramadol Hydrochloride Injection is mixed in the same syringe with injections of diazepam, diclofenac sodium, indomethacin, midazolam and piroxicam.

SHELF LIFE:

Unopened (Before Reconstitution): 36 months from the date of manufacture.

After Reconstitution:

Reconstituted solution is physically, chemically and microbiologically compatible with following infusion solutions, in PVC bag and Type-1 glass bottle, when stored at 2°C-8°C for 5 days:

- 0.9% Sodium Chloride Intravenous Infusion
- 5% Glucose Intravenous infusion
- Sodium Lactate Solution
- Haemaccel

SPECIAL PRECAUTIONS FOR STORAGE:

Unopened (Before Reconstitution): Store below 30° C. Protect from light. Keep medicine out of reach of children

After Reconstitution:

In keeping with good pharmaceutical practice, IV solutions should be prepared at the time of infusion, under the appropriate aseptic conditions.

Reconstituted solution is physically, chemically and microbiologically compatible with following infusion solutions, in PVC bag and Type-1 glass bottle, when stored at 2°C-8°C for 5 days:

- 0.9% Sodium Chloride Intravenous Infusion
- 5% Glucose Intravenous infusion
- Sodium Lactate Solution
- Haemaccel

INSTRUCTIONS FOR USE/HANDLING:

The solution for injection is unpreserved, should only be used once and injected or diluted immediately after opening. Any remaining solution should be discarded.

In keeping with good pharmaceutical practice, IV solutions should be prepared at the time of infusion, under the appropriate aseptic conditions.

Compatibility with IV fluids

Reconstituted solution is physically, chemically and microbiologically compatible with following infusion solutions, in PVC bag and Type-1 glass bottle, when stored at 2°C-8°C for 5 days:

- 0.9% Sodium Chloride Intravenous Infusion
- 5% Glucose Intravenous infusion
- Sodium Lactate Solution
- Haemaccel

DOSAGE FORMS AND PACKAGING AVAILABLE:

TRAMASURE-100: Clear transparent glass ampoules having 2ml capacity is packed in clear ALU/PVC blister. 4 such ampoules in clear ALU/PVC blister are packed in a carton along with a package insert.

DATE OF REVISION: Sep'2022



Manufactured by:

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